Professional Installation Manual

Enterprise Server Deployment, Virtualization, and Administration Standard

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1. Introduction & Purpose

This manual provides an authoritative, step-by-step technical reference designed for system administrators deploying production-grade Ubuntu Server instances within virtualized environments. Following this guide ensures consistent deployment architectures, standardized hostname declarations (`server01`), hardened security postures, and verified network interconnectivity.

2. Hardware & Virtualization Requirements

Before initiating provisioning, verify host hardware compatibility and allocate the following minimum resource profile within Oracle VM VirtualBox:

Resource Component	Allocation Specification	Operational Purpose
Hypervisor	Oracle VM VirtualBox (Latest)	Host virtualization engine
RAM	4096 MB (4 GB)	Guarantees stable multi-service uptime
Processor (CPU)	2 Virtual Cores	Optimizes execution of kernel and daemons
Storage	40 GB Virtual Disk	Accommodates root FS and logs

3. Ubuntu Server Installation Guide

1. **Virtual Machine Creation:** Open VirtualBox, create a new VM titled `server01`, allocate 4GB RAM, 2 CPUs, and attach the official Ubuntu Server LTS ISO image to the optical drive.
2. **Storage Provisioning:** Configure a 40 GB dynamically allocated virtual hard disk using the VDI format.
3. **Installer Setup:** Boot the virtual machine. Select your preferred system language, keyboard layout, and choose the standard Ubuntu Server installation profile.
4. **Network & Storage Layout:** Accept default DHCP network settings on the primary interface (`enp0s3`) and select guided storage layout utilizing the entire virtual disk.
5. **User Profile Creation:** Enter administrative credentials, specifying username `vboxuser` and a strong master passphrase.
6. **OpenSSH Integration:** *Critical Step:* Toggle the option to install the OpenSSH server package during the installer's software selection prompt.

4. Initial Server Configuration

Once installation finishes and the system reboots, execute the following administrative configuration tasks:

- **Verify Hostname:** Confirm the system hostname is active as `server01`. If modification is required, update `/etc/hostname` and `/etc/hosts`.
- **Package Repository Refresh:** Update local package indexes to ensure software source integrity: `sudo apt update && sudo apt upgrade -y`.
- **Firewall Configuration:** Enable Uncomplicated Firewall (UFW) and allow secure remote shell traffic: `sudo ufw allow OpenSSH && sudo ufw enable`.

5. Server Verification Procedures

Execute the following diagnostic sequence to validate operational readiness:

> `hostname` (Verify hostname is `server01`)
> `ip addr` (Confirm network interface `enp0s3` and IP address)
> `ping -c 4 8.8.8.8` (Verify gateway and internet connectivity)
> `systemctl status ssh` (Confirm OpenSSH daemon is active and running)

6. Troubleshooting Guide

- **Missing SSH Service:** If verification returns `Unit ssh.service could not be found`, install it manually via `sudo apt install openssh-server -y` and enable it with `sudo systemctl enable --now ssh`.
- **Network Interface Down:** If `ip addr` shows no dynamic IP, verify VirtualBox network adapter settings are set to NAT or Bridged Adapter and restart networking.

7. Best Practices for Administrators

- Always enforce SSH key-based authentication in production environments and disable root password logins.
- Maintain regular snapshots in VirtualBox prior to applying major kernel upgrades or package distributions.
- Document all custom configurations and port allocations inside the repository `README.md`.